



PROFILE

Junior Computer Engineer with a strong background in software development, data engineering, Business Intelligence, machine learning, and decision-support systems. Experienced in designing web platforms, building data pipelines, developing analytical dashboards, and delivering intelligent solutions that transform operational data into actionable insights.

EDUCATION

Engineering Degree – Computer Science, MIAGE	2023 – 2026
EMSI, Tangier	
Master’s Degree – Artificial Intelligence for Digital Economy and Management	2024 – 2026
Faculty of Legal, Economic and Social Sciences, Tangier	
Bachelor’s Degree – Mathematical Sciences and Applications	2020 – 2023
Faculty of Sciences, Tetouan	

EXPERIENCE

Final-Year Project – Intelligent Decision-Support System for Industrial Performance	Feb. – Aug. 2026
LELMA, Mecachrome Group, Tangier	
- Designed an intelligent decision-support system for industrial performance monitoring, covering production, downtime, objectives, and quality defect data. - Developed a multi-UAP web platform with role-based access control, data validation workflows, and dashboards tailored to different user profiles. - Built a data engineering pipeline to clean, standardize, and centralize Excel and Power BI data into DuckDB. - Developed a Streamlit supervision application for interactive KPI monitoring, production forecasting, low-performance risk detection, and quality defect analysis. - Maintained and enhanced existing Power BI reports and integrated them into the project ecosystem.	
Technologies: Python, Django, MySQL/MariaDB, DuckDB, Power BI, DAX, Power Query, Streamlit, Plotly, pandas, Airflow, Docker, scikit-learn, Prompt Engineering, HTML, CSS, JavaScript.	

Advanced Internship	Jul. – Sep. 2025
HB Development, Tetouan	
- Developed an interactive educational platform based on a service-oriented architecture. - Implemented backend APIs, authentication mechanisms, and deployment workflows using modern development tools.	
Technologies: MERN Stack, FastAPI, Docker, Git, JWT, Postman, Proxmox, NLP.	

Internship	Jul. – Aug. 2024
Amendis (Veolia), Tetouan	
- Developed an internal web application for IT infrastructure and network equipment management. - Designed the MySQL database and implemented CRUD, search, and consultation features.	
Technologies: Python, Flask, MySQL, Bootstrap.	

PROJECTS

Cognitive Twin for Developers
AI-powered software assistant for code analysis, developer style learning, automated code generation, debugging, and UML generation using Next.js, FastAPI, PostgreSQL, pgvector, Tree-sitter, and Ollama.
Real-Time Big Data Pipeline – E-commerce
Distributed streaming pipeline for e-commerce transaction analysis using Kafka, Spark Streaming, Spark SQL, Scala, Docker, and Parquet.
BI Data Integration Project
ETL pipeline and Power BI dashboards for analytical reporting and business performance monitoring using SQL Server, SSIS, DAX, and Data Warehouse concepts.
J2EE Commercial Management System
Sales, inventory, and price management system based on a microservices architecture using Spring Boot, JPA, JWT, REST APIs, and MySQL.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, C, C++
Data Engineering & BI: Power BI, DAX, Power Query, ETL, Data Warehousing, KPI Modeling, DuckDB, pandas, Plotly
ML & DL: scikit-learn, TensorFlow/Keras, PyTorch, NLP, Prompt Engineering, Feature Engineering, Model Evaluation
Web & Backend Development: Django, FastAPI, Spring Boot, React, Streamlit, REST APIs, HTML, CSS
Databases: MySQL/MariaDB, PostgreSQL, MongoDB, Firebase, Cassandra, DuckDB
Tools & Methodologies: Git, Docker, Airflow, Linux, Proxmox, Hadoop, Postman, SCRUM, UML, Merise, DevOps

CERTIFICATIONS

IBM AI Engineering (Ongoing) – Meta Django – Meta React / React Native – Unix Workbench – Azure Networks

LANGUAGES

Arabic: Native French: Advanced English: Advanced